FAST — National University of Computer and Emerging Sciences Karachi Campus

Department of Software Engineering **PF LAB TASKS** **Course Code**: C1002

Subject: Programming Fundamentals

Section: D & C

Instructor: Mr. Shayan Ali

STUDENT SUBMISSION DETAILS

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Class / Section: C

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In Lab Tasks

Question 1: Shopkeeper Billing System (2D Array)

A shopkeeper sells 5 different products. Write a C program using a 2D array to store the quantity and unit price of each product. Calculate the total price of each product using quantity × price, display the quantity, unit price, and total price of each product, and calculate and display the grand total of all products using nested loops.

[*] Untitled1

```c
#include <stdio.h>

int main()
{
    int a[5][2];
    int i, j;
    int total;
    int grandTotal = 0;

    for(i = 0; i < 5; i++)
    {
        printf("Enter quantity and unit price of product %d: ", i + 1);

        for(j = 0; j < 2; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("\nProduct\tQuantity\tPrice\tTotal\n");

    for(i = 0; i < 5; i++)
    {
        total = a[i][0] * a[i][1];
        grandTotal = grandTotal + total;

        printf("%d\t%d\t\t%d\t%d\n",
               i + 1, a[i][0], a[i][1], total);
    }

    printf("\nGrand Total = %d\n", grandTotal);

    return 0;
}
```

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Compiler Resources Compile Log Debug Find Results Console

C:\Users\User\Desktop\Untitled1.exe

```
Enter quantity and unit price of product 1: 1 150
Enter quantity and unit price of product 2:  4 780
Enter quantity and unit price of product 3:  1 890
Enter quantity and unit price of product 4: 2 450
Enter quantity and unit price of product 5: 4 690

Product Quantity        Price   Total
1       1               150     150
2       4               780     3120
3       1               890     890
4       2               450     900
5       4               690     2760

Grand Total = 7820

--------------------------------
Process exited after 66.47 seconds with return value 0
Press any key to continue . . .
```

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Question 2: Print the Following Pattern

Write a C program using nested loops to print the following pattern:

1

1 4

1 4 27

1 4 27 16

1 4 27 16 125

[*] Untitled1.cpp

```c
#include <stdio.h>

int main()
{
    int i, j;
    int num;

    for(i = 1; i <= 5; i++)
    {
        for(j = 1; j <= i; j++)
        {
            if(j == 1)
                num = 1;
            else if(j == 2)
                num = 4;
            else if(j == 3)
                num = 27;
            else if(j == 4)
                num = 16;
            else
                num = 125;

            printf("%d ", num);
        }

        printf("\n");
    }

    return 0;
}
```

```
1
1 4
1 4 27
1 4 27 16
1 4 27 16 125

--------------------------------
Process exited after 0.0714 seconds with return value 0
Press any key to continue . . .
```

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Question 3: Print the Following Pattern

Write a C program using nested loops to print the following pattern:

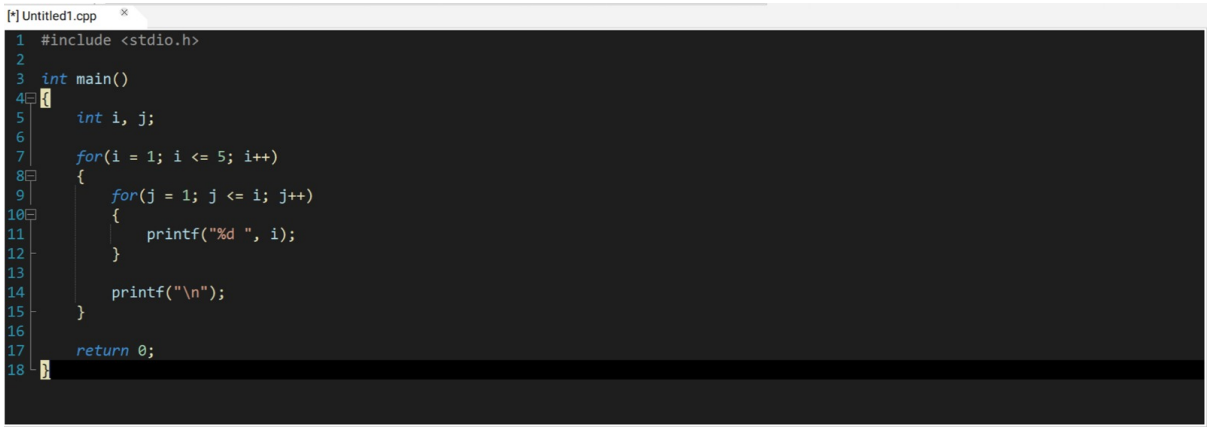
1

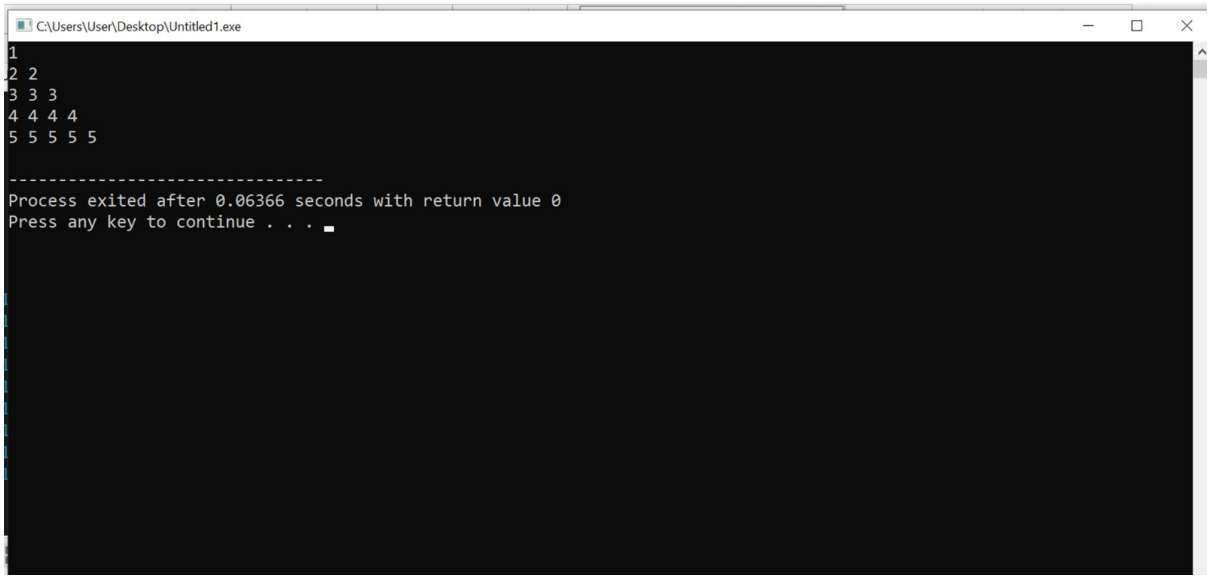
2 2

3 3 3

4 4 4 4

5 5 5 5 5





Question 4: Print the Following Pattern

Write a C program using nested loops to print the following pattern:

1 2 3 4 5

1 2 3 4

1 2 3

1 2

1

[*] Untitled1.cpp

```c
#include <stdio.h>

int main()
{
    int i, j;

    for(i = 5; i >= 1; i--)
    {
        for(j = 1; j <= i; j++)
        {
            printf("%d ", j);
        }

        printf("\n");
    }

    return 0;
}
```



Question 5: 3×3 Matrix Types Using 2D Arrays in C

Write a C program that takes input for a 3×3 matrix using a 2D array. Apply appropriate conditions using nested loops to check and display whether the matrix satisfies the following types:

1. Square Matrix
2. Null Matrix
3. Diagonal Matrix
4. Upper Triangular Matrix
5. Lower Triangular Matrix
6. Identity Matrix

Display all applicable matrix types. A single matrix may satisfy multiple conditions. At the end, using loops, also display the diagonal elements, the upper triangular

[*] Untitled1.cpp

```c
#include <stdio.h>

int main()
{
    int a[3][3];
    int i, j;

    int square = 1;
    int null = 1;
    int diagonal = 1;
    int upper = 1;
    int lower = 1;
    int identity = 1;

    printf("Enter the 9 elements of the matrix:\n");

    for(i = 0; i < 3; i++)
    {
        for(j = 0; j < 3; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("\nMatrix:\n");
```

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Untitled1.cpp

```c
        for(j = 0; j < 3; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("\nMatrix:\n");

    for(i = 0; i < 3; i++)
    {
        for(j = 0; j < 3; j++)
        {
            printf("%d ", a[i][j]);
        }
        printf("\n");
    }

    /* Check matrix types */

    for(i = 0; i < 3; i++)
    {
        for(j = 0; j < 3; j++)
        {
            if(a[i][j] != 0)
                null = 0;
```

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```c
    for(i = 0; i < 3; i++)
    {
        for(j = 0; j < 3; j++)
        {
            if(a[i][j] != 0)
                null = 0;

            if(i != j && a[i][j] != 0)
                diagonal = 0;

            if(i > j && a[i][j] != 0)
                upper = 0;

            if(i < j && a[i][j] != 0)
                lower = 0;

            if(i == j && a[i][j] != 1)
                identity = 0;

            if(i != j && a[i][j] != 0)
                identity = 0;
        }
    }

    /* Display applicable types */
```

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```
38      for(i = 0; i < 3; i++)
39      {
40          for(j = 0; j < 3; j++)
41          {
42              if(a[i][j] != 0)
43                  null = 0;
44
45              if(i != j && a[i][j] != 0)
46                  diagonal = 0;
47
48              if(i > j && a[i][j] != 0)
49                  upper = 0;
50
51              if(i < j && a[i][j] != 0)
52                  lower = 0;
53
54              if(i == j && a[i][j] != 1)
55                  identity = 0;
56
57              if(i != j && a[i][j] != 0)
58                  identity = 0;
59          }
60      }
61
62      /* Display applicable types */
```

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*] Untitled1.cpp

```
83
84      /* Display diagonal elements */
85
86      printf("\nDiagonal Elements: ");
87
88      for(i = 0; i < 3; i++)
89      {
90          printf("%d ", a[i][i]);
91      }
92
93      /* Display upper triangular elements */
94
95      printf("\n\nUpper Triangular Elements:\n");
96
97      for(i = 0; i < 3; i++)
98      {
99          for(j = 0; j < 3; j++)
100         {
101             if(i <= j)
102                 printf("%d ", a[i][j]);
103             else
104                 printf("  ");
105         }
106
107         printf("\n");
```

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```
104                 printf("  ");
105         }
106
107         printf("\n");
108     }
109
110     /* Display lower triangular elements */
111
112     printf("\nLower Triangular Elements:\n");
113
114     for(i = 0; i < 3; i++)
115     {
116         for(j = 0; j < 3; j++)
117         {
118             if(i >= j)
119                 printf("%d ", a[i][j]);
120             else
121                 printf("  ");
122         }
123
124         printf("\n");
125     }
126
127     return 0;
128 }
```

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